

AASD 4010

Deep Learning - I

Applied AI Solutions Developer Program



NER



Agenda



Named Entity Recognition
NER format
NER architecture
Bidirectional LSTM
CRF
Dataturks



Named Entity Recognition

NER

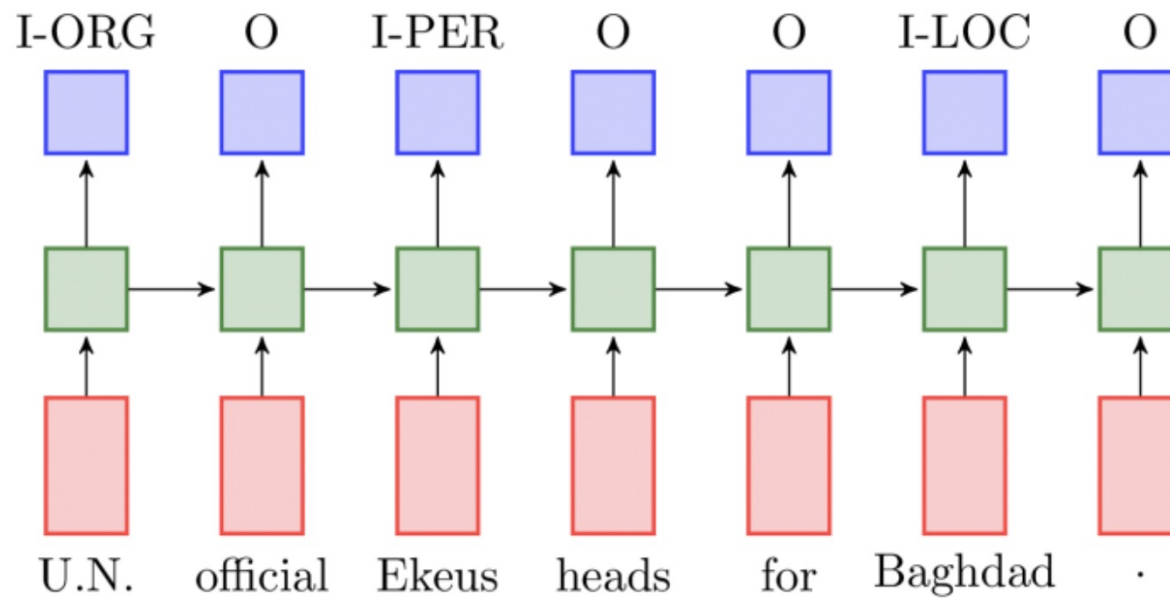


NER

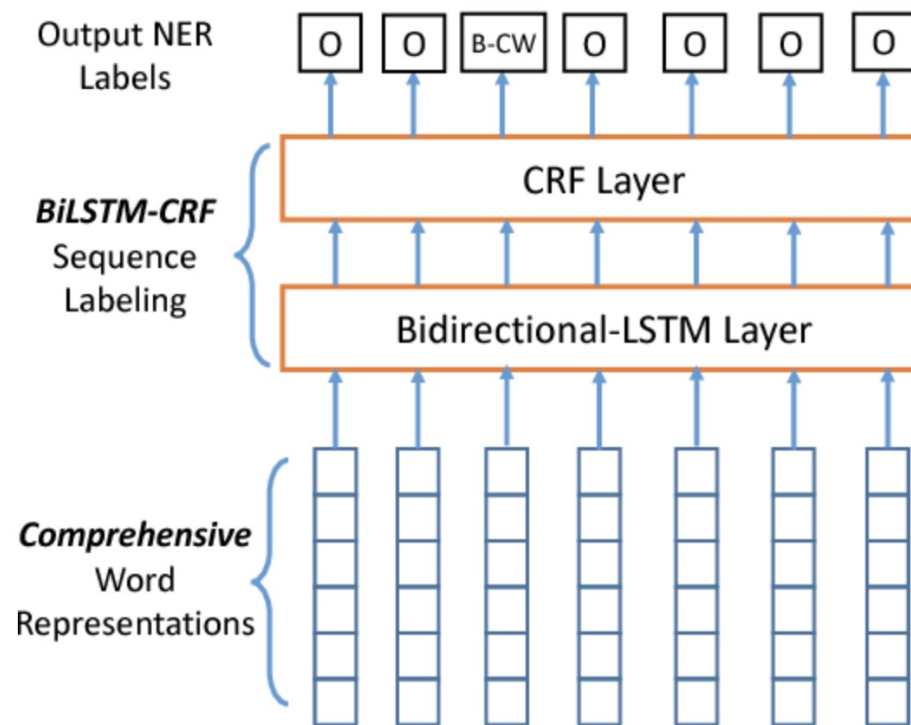
F.B.I. Agent **Peter Strzok PERSON**, **Who Criticized Trump PERSON** in Texts, Is **Fired GPE** - **The New York Times ORG** SectionsSEARCHSkip to contentSkip to site indexPoliticsSubscribeLog InSubscribeLog InToday's **PaperAdvertisementSupported ORG** byF.B.I. Agent **Peter Strzok PERSON**, **Who Criticized Trump PERSON** in Texts, Is FiredImagePeter Strzok, a top **F.B.I. GPE** counterintelligence agent who was taken off the special counsel investigation after his disparaging texts about President **Trump PERSON** were uncovered, was fired. **CreditT.J. Kirkpatrick PERSON** for **The New York TimesBy Adam Goldman ORG** and **Michael S. SchmidtAug PERSON**. **13 CARDINAL**, **2018WASHINGTON CARDINAL** — **Peter Strzok PERSON**, the **F.B.I. GPE** senior counterintelligence agent who disparaged President **Trump PERSON** in inflammatory text messages and helped oversee the **Hillary Clinton PERSON** email and **Russia GPE** investigations, has been fired for violating bureau policies, Mr. **Strzok PERSON**'s lawyer said **Monday DATE**. Mr. Trump and his allies seized on the texts — exchanged during the **2016 DATE** campaign with a former **F.B.I. GPE** lawyer, **Lisa Page — in PERSON** assailing the **Russia GPE** investigation as an illegitimate "witch hunt." Mr. **Strzok PERSON**, who rose over **20 years DATE** at the **F.B.I. GPE** to become one of its most experienced counterintelligence agents, was a key figure in **the early months DATE** of the inquiry. Along with writing the texts, Mr. **Strzok PERSON** was accused of sending a highly sensitive search warrant to his personal email account. The **F.B.I. GPE** had been under immense political pressure by Mr. **Trump PERSON** to dismiss Mr. **Strzok PERSON**, who was removed **last summer DATE** from the staff of the special counsel, **Robert S. Mueller III PERSON**. The president has repeatedly denounced Mr. **Strzok PERSON** in posts on **Twitter EVENT**, and on **Monday DATE** expressed satisfaction that he had been sacked. Mr. **Trump's ORG** victory traces back to **June DATE**, when Mr. **Strzok PERSON**'s conduct was laid out in a wide-ranging inspector general's report on how the **F.B.I. GPE** handled the investigation of **Hillary Clinton's PERSON** emails in the run-up to the **2016 DATE** election. The report was critical of Mr. **Strzok PERSON**'s conduct in sending the



NER format



NER architecture



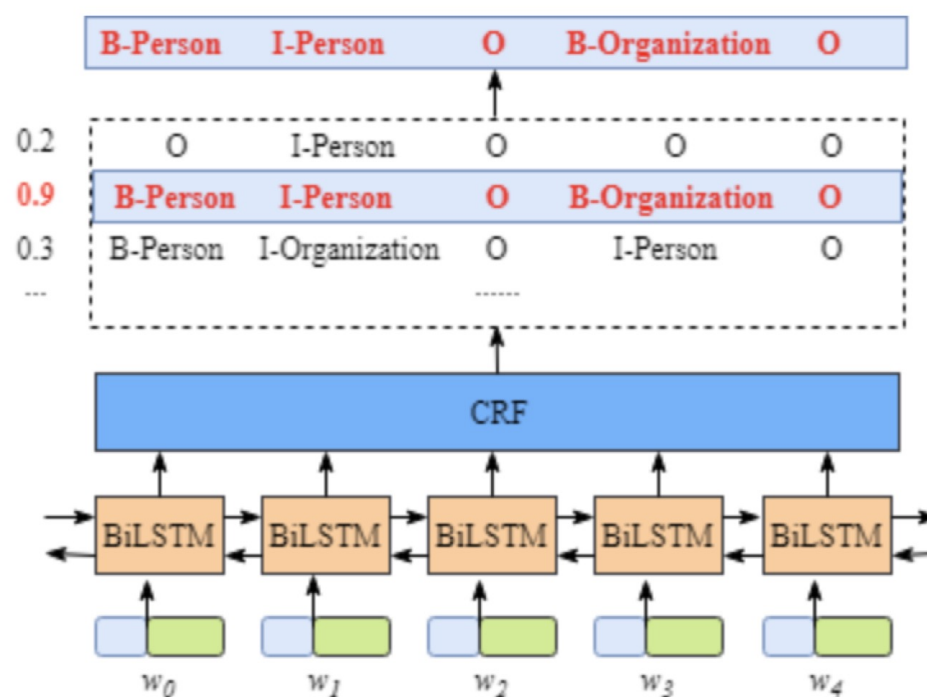
NER architecture

First, every word in sentence X is represented as a vector which includes the word's character embedding and word embedding. The character embedding is initialized randomly. The word embedding usually is from a pre-trained word embedding file. All the embeddings will be fine-tuned during the training process.

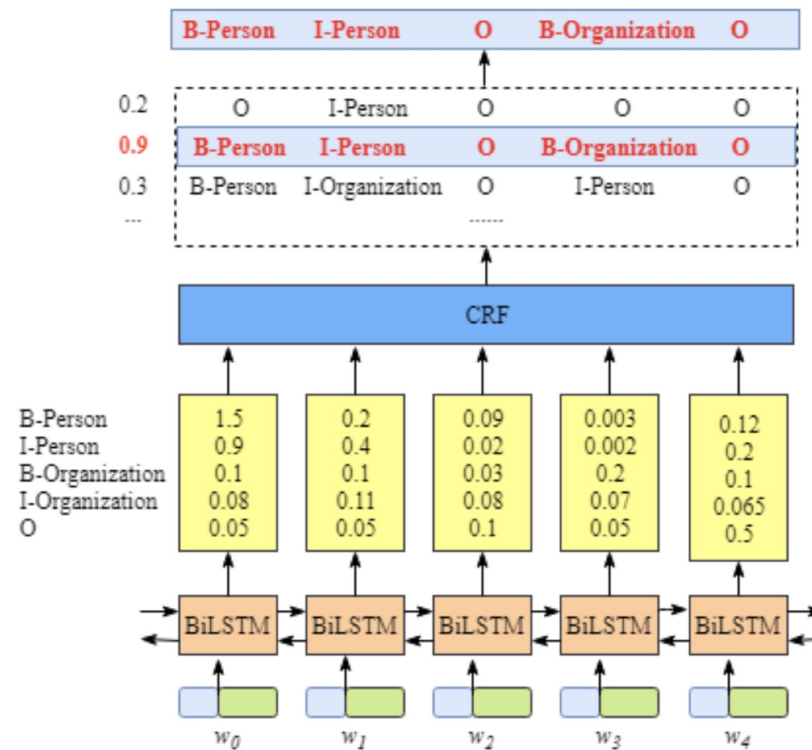
Second, the inputs of BiLSTM-CRF model are those embeddings and the outputs are predicted labels for words in sentence X .



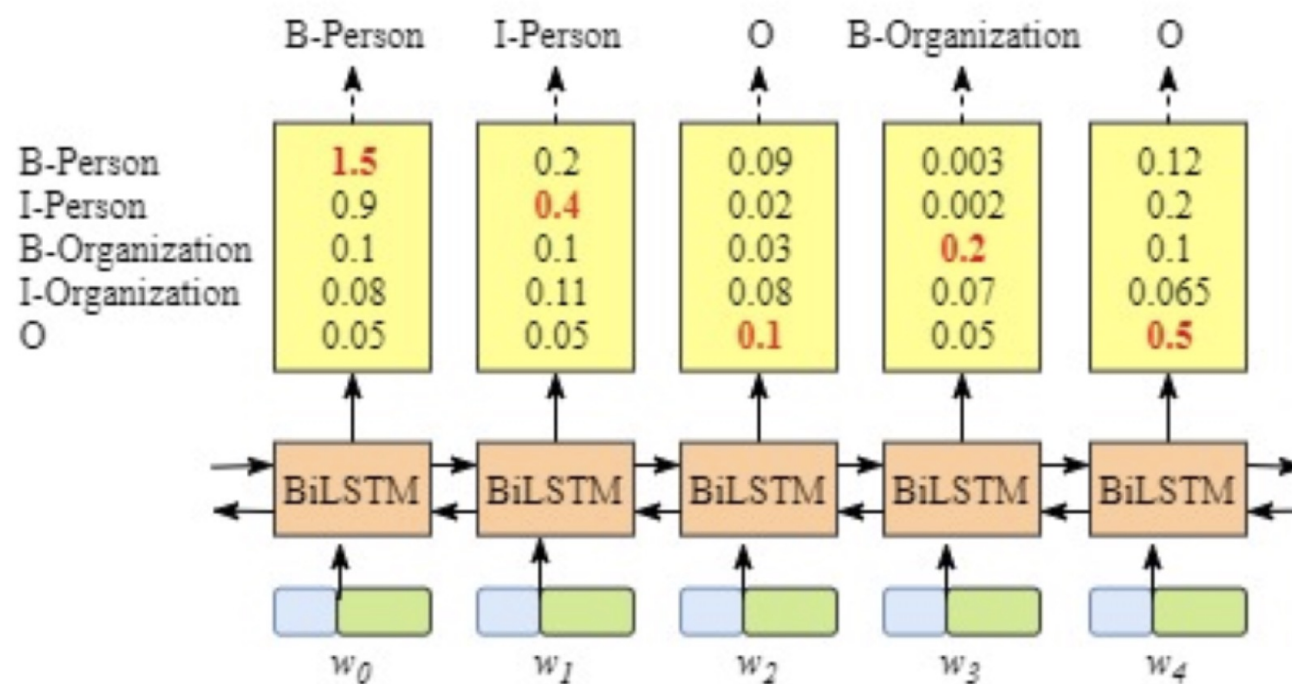
NER architecture



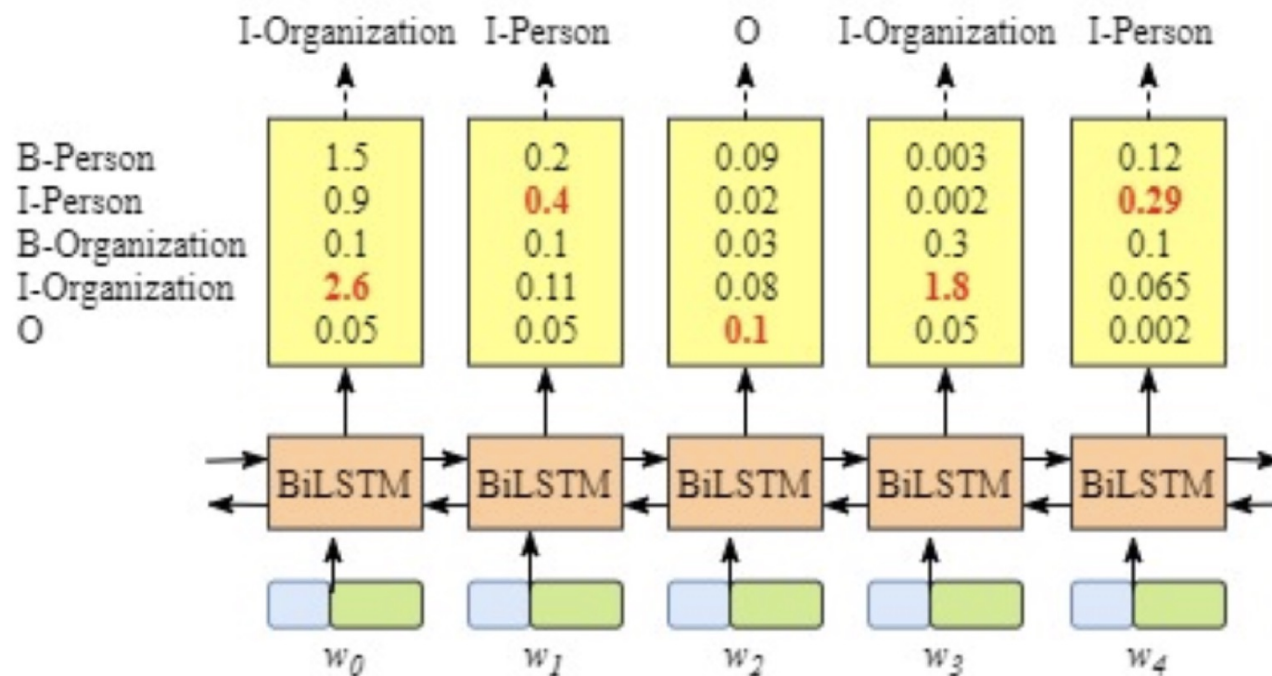
NER architecture



No CRF layer



No CRF layer



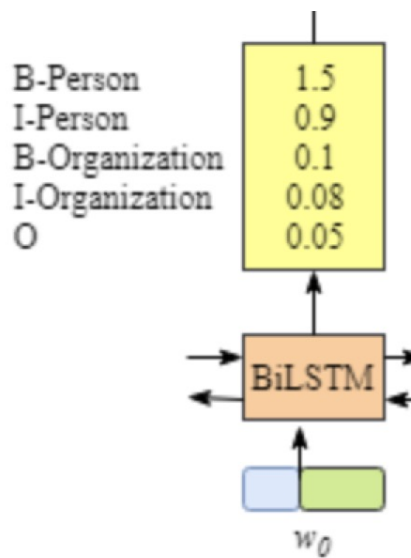
CRF learns CONSTRAINTS

- The label of the first word in a sentence should start with “B-“ or “O”, not “I-“
- “B-label1 I-label2 I-label3 I-...”, in this pattern, label1, label2, label3 ... should be the same named entity label. For example, “B-Person I-Person” is valid, but “B-Person I-Organization” is invalid.
- “O I-label” is invalid. The first label of one named entity should start with “B-“ not “I-“, in other words, the valid pattern should be “O B-label”
- ...



CRF Scores

Emission Score



Transition Score

	START	B-Person	I-Person	B-Organization	I-Organization	O	END
START	0	0.8	0.007	0.7	0.0008	0.9	0.08
B-Person	0	0.6	0.9	0.2	0.0006	0.6	0.009
I-Person	-1	0.5	0.53	0.55	0.0003	0.85	0.008
B-Organization	0.9	0.5	0.0003	0.25	0.8	0.77	0.006
I-Organization	-0.9	0.45	0.007	0.7	0.65	0.76	0.2
O	0	0.65	0.0007	0.7	0.0008	0.9	0.08
END	0	0	0	0	0	0	0



CRF learns CONSTRAINTS

The label of the first word in a sentence should start with “B-“ or “O”, not “I-“
(The transition scores from “START” to “I-Person or I-Organization” are very low.)

“B-label1 I-label2 I-label3 I-...”, in this pattern, label1, label2, label3 ... should be the same named entity label. For example, “B-Person I-Person” is valid, but “B-Person I-Organization” is invalid.

(Score from “B-Organization” to “I-Person” is only 0.0003 much lower than the others.)

“O I-label” is invalid. The first label of one named entity should start with “B-“ not “I-“, in other words, the valid pattern should be “O B-label”

(Again, the score from “O” to “I-Person” is very small.)



Problem



Problem

Problem: Create a NER system

Dataset:

<https://www.kaggle.com/abhinavwalia95/entity-annotated-corpus>



Dataset

<https://www.kaggle.com/abhinavwalia95/entity-annotated-corpus>

	Sentence #	Word	POS	Tag
0	Sentence: 1	Thousands	NNS	O
1	NaN	of	IN	O
2	NaN	demonstrators	NNS	O
3	NaN	have	VBP	O
4	NaN	marched	VBN	O
5	NaN	through	IN	O
6	NaN	London	NNP	B-geo
7	NaN	to	TO	O
8	NaN	protest	VB	O
9	NaN	the	DT	O
10	NaN	war	NN	O



Steps

<https://www.kaggle.com/abhinavwalia95/entity-annotated-corpus>

Import all the necessary libraries

Read the dataset

Split them into sentences

Convert the words, tags into IDs

Find Maximum length of Input Sentences

Pad Input Sequences and tags with maximum input length

Split the data into training and testing set

Create Bidirectional LSTM layers

Create CRF Layer

Create the model

Compile the model

Fit the model

Plot the learnings



Dataturks



Daturks

Data labeling platform for data scientists to label data in a collaborative manner.

It **was** accessed using daturks.com (web interface). Not anymore

Access Steps:

Install docker in your local system

Create account in dockerhub.com

Pull the docker image from dockerhub into your local machine

Run the downloaded docker image

Access the daturks portal using localhost (web browser)



Datururks access through Docker Image

Install docker in your local system

Create account in dockerhub.com

Pull the docker image from dockerhub into your local machine

```
docker pull cprakashagr/datururks:3.3.0
```

Run the downloaded docker image

```
docker run -d -p 80:80 cprakashagr/datururks:3.3.0
```

Access the datururks portal using localhost (web browser)



Daturks Homepage

Daturks

Documentation APIs Try Demo Logout

Subash

Home

Create Dataset

Get API Key

Onprem users

Labels Used

0

Total Labels in Package

2500000

Datasets

Sample

Demo Image Polygon Bounding Box

Demo Image Bounding Box Project

Demo Document Annotations

My Projects

Refresh

Owner

Sample

PRIVATE

Document Annotation

100%

1 / 1

Overview

Demo Image Polygon Bounding Box

PUBLIC

Image Segmentation

20%

10 / 50

Overview

Demo Image Bounding Box Project

PUBLIC

Image Segmentation

20%

10 / 50

Overview

Demo Document Annotations

PUBLIC

Document Annotation

18%

9 / 49

Overview

Daturks Blog Contact Privacy Policy

(+91) 080-331-72755, +91-99010-49915, +91-88614-08222

contact@daturks.com



Choosing a Project type

Daturks

Documentation APIs Try Demo Logout

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Labels Used

0

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2500000

Datasets

Sample

Demo Image Polygon Bounding Box

Demo Image Bounding Box Project

Demo Document Annotations

Create Dataset

How to create a dataset? [See Demo Videos](#)

Image Annotations

Image Classification

Image Classification

Upload file with image urls and get your team to classify image.

Create Dataset

Image Bounding Box

Image Bounding Box

Create rectangular bounding boxes around objects in Images.

Create Dataset

Image Segmentation

Bounding Box, Polygons etc.

Create free-form polygons, points, lines in images.

Create Dataset

Video Annotations

Video Classification

Video Classification

Upload file with video urls and get your team to classify videos.

Create Dataset

Video Annotation

Video Annotation

Object Tracking: Create rectangular bounding boxes around objects in video.

Create Dataset



Choosing a Project type

Text Annotations

Document Annotation

Document Processing, NLP

Full-length document NER annotation. Annotate individual files (PDF, DOC, Text) one-by-one

Create Dataset

NER Tagging

NLP, Text Processing, Large Sentences

Line-by-line NER annotation. Upload a CSV file and do NER on each row.

Create Dataset

PoS Tagging for Small Sentences

NLP, Text Processing, Search Queries and Logs

Optimized for small sentences. Annotate words in a sentence.

Create Dataset

Text Summarization

NLP, Text Processing

Upload file with text and invite people to write summaries

Create Dataset

Text Classification

NLP, Text Processing, IR

Create project with entities and text, get your team to classify them

Create Dataset

Text Moderation

Text Processing, User Generated Content

Get your team to moderate the user generated text

Create Dataset

Custom Annotations

Email us for Custom Annotations

If there is any annotation use-case which is not covered in the list, please drop us an email at contact@dataturks.com



Creating a Dataset Project

Daturks

Documentation APIs Try Demo Logout

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Home

Create Dataset

Get API Key

Onprem users

Labels Used

0

Total Labels in Package

2500000

Datasets

Sample

Demo Image Polygon Bounding Box

Demo Image Bounding Box Project

Demo Document Annotations

Create Dataset

Dataset Name

Dataset name

List of Entities/Categories

List of Classes comma seperated : Car, Cat, Tree

Tagging Instruction

Document annotation guidelines for your team. e.g Mark javascript as super-human skill

Advanced Options

Add Classification

Submit



Creating a Dataset Project

Create Dataset

Dataset Name

SampleNER

List of Entities/Categories

Drug, Dose, Disease, AdverseEvent

Tagging Instruction

Tag all the entities in the document

► Advanced Options

• Add Classification

Submit



Upload data to the project

Select Upload Type

Upload Raw Data

Upload Pre-Annotated Data

If you have some data which is already pre-annotated and want to go through annotations and correct them.



Upload data to the project

Import Data

Change Import Type

Select file with input data

Please upload a valid text/doc/pdf file.

OR

A zip file of all the text/pdf/doc documents. Max file size is 100 MB for free plans

► [Advanced Details](#)

Choose File [No file chosen](#)

Submit



Upload data to the project

Project creation successful

1 NUMBER OF HITS CREATED

6.09KB FILE SIZE

Start Tagging



Document uploaded

←

🏠

Project Guidelines

Not Done

☒ AutoClose on Selection

☒ Autolabel Same Text in Document

Undo

Clear All

🗨

✎

• Click on the document and then drag to select text and select a label.

• More queries? [See Demo Videos](#)

Entities

Drug

Disease

Chemical

AdverseEvent

Nubeqa (darolutamide) is an androgen receptor inhibitor indicated for the treatment of patients with non-metastatic castration-resistant prostate cancer.

Pharmacokinetics of darolutamide were determine as a part of the phase-2 study, which determined dose escalation, dose expansion, and long term follow-up.

Pharmacokinetics were determined in patients receiving 200 mg/day (n=3), 400 mg/day (n=7), 600 mg/day (n=3) 1000 mg/day (n=3), 1400 mg/day (n=3), 1800 mg/day (n=3) with food.

No intra-patient escalation was permitted.

Pharmacokinetic analysis of darolutamide showed rapid absorbed and the median t max ranged 3.0-5.1 hours for darolutamide and 1.5-5.0 h for diastereo isomer keto-darolutamide on day 1.

The mean metabolite to parent ratio (area under the curve, AUC) at steady state was 1.6-2.3:1.

The exposure of darolutamide (AUC t a nd C max) at steady state increased in a linear, dose-related manner up to 1400 mg daily dose.

Most common Grade 3 adverse events occurring in > 10% of patients were fatigue or asthenia 2 (2%), back pain 2 (2%) and arthralgia 2 (2%).

An open randomized phase 1 study evaluated safety and tolerability, bioavailability and effect of food on bioavailability of darolutamide in patients with metastatic chemotherapy naïve CRPC.

The study compared relative bioavailability of 600 mg tablets (tablet A and tablet B) and capsule (reference) formulations in patients with chemotherapy naïve metastatic castrate resistant prostate cancer.

Patients were randomized to one of 3 treatment sequences in each group.

There were three treatment periods for each patient during the PK Component which included administration of two darolutamide 300 mg test tablet (Tab A or Tab B) in both fasted and fed conditions, or administration of darolutamide 100 mg capsule at 600 mg dose in fed state.

There was at least 7 days wash-out period between the three study treatment administrations.

The total daily dose was 600 mg/day

Thirty men with were enrolled. Area under the concentration curve (AUC) and maximum concentration (C max) values were approximately two times lower after administration at fast. With high fat meal the C max occurred about 2-3 hours later.

The food effect was similar in all groups.

The pharmacokinetics observed with the two new tablet formulations were comparable with the pharmacokinetics of the capsule formulation

Nubeqa (darolutamide) is an androgen receptor inhibitor indicated for the treatment of patients with non-metastatic castration-resistant prostate cancer.

Phase 1-2 study

Phase 1-2 clinical trial was conducted to assess darolutamide in men with progressive metastatic castration-resistant prostate cancer.

The primary objectives of the trial were to assess safety and tolerability and to define a maximum tolerated dose. Secondary objectives were to determine the pharmacokinetics, and antitumor activity as defined by changes in serum prostate-specific antigen (PSA), by imaging of soft tissue and bone lesions, and by changes in circulating tumor cell counts.

Tolerability and outcome measures are discussed elsewhere.

Blood samples were collected for the pharmacokinetics analyses.

We identified plasma concentrations of diastereoisomers (S,R)-darolutamide and (S,S)-darolutamide, and the major metabolite keto-darolutamide by liquid chromatography-tandem mass spectrometry (LC-MS/MS):



Tagging / Labeling

Entities

Drug Dose Disease AdverseEvent

Nubeqa (darolutamide) is an androgen receptor inhibitor indicated for the treatment of metastatic castration-resistant prostate cancer. Pharmacokinetics of darolutamide were determined as a part of the phase II study. Pharmacokinetics were determined in patients receiving 200 mg/day (n=3), 1800 mg/day (n=3), and 1800 mg/day (n=3). No intra-patient escalation was permitted. Pharmacokinetic analysis of darolutamide showed rapid absorption and the mean metabolite to parent ratio (area under the curve, AUC) at steady state was 1.0-2.5:1. The exposure of darolutamide (AUC t₀₋₂₄ and C_{max}) at steady state increased in a linear, dose-related manner up to 1400 mg daily dose.

Save (enter) (esc) X

Nubeqa

- Drug
- Dose
- Disease
- AdverseEvent



Tagging / Labeling

Drug
Dose
Disease
AdverseEvent

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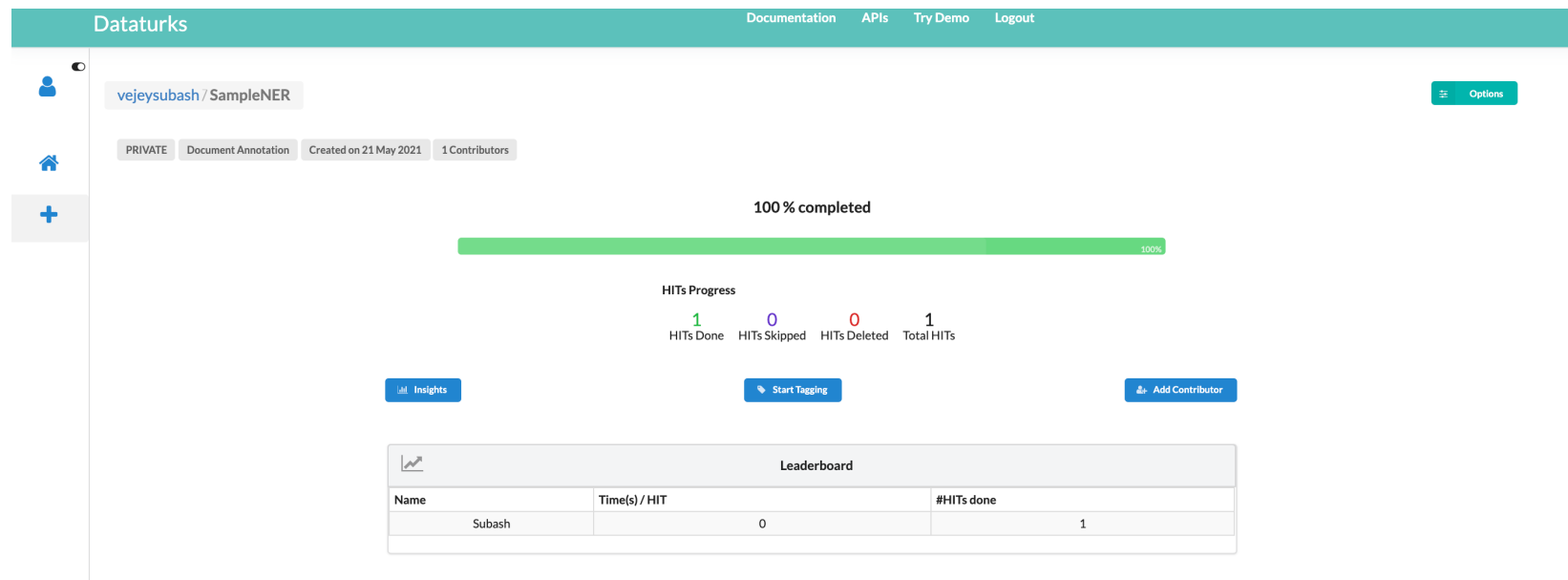
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Tagging / Labeling (Finished)



Download the labeled data

vejeysubash / SampleNER

Export Data

☐ All Items ☒ Completed Items

Download file would be a text file where each line is a JSON containing the selected text, start index, end index and marked category

☒ Json Format ☐ Stanford NER Format

Download file

Convert to Spacy format

Options

- + Add Data
- Edit Project
- HITs Done
- HITs Skipped
- Download
- Keyboard Shortcuts
- Contributors
- Delete Project



Further Reading

NER

https://keras.io/getting_started/intro_to_keras_for_engineers/

Dataturks

<https://github.com/DataTurks/DataTurks>

Assignment Reading

<https://github.com/DataTurks/Entity-Recognition-In-Resumes-SpaCy>

Dataset for Assignment

Resume dataset

